

Shift Factors on the Irish Transmission Grid

The DC pipeline in full: K , B , L , L^+ , PTDF, sensitivity

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Network: WP2033_all-island 754 buses, 980 branches, 157 wind nodes

1 Shift factor, sensitivity, PTDF

The shift factor, as it is used today

For a branch e and a generator at node n , the shift factor is the change in flow on e per MW of injection change at n :

$$\text{SF}_{e,n} = \frac{\Delta f_e}{\Delta P_n}. \quad (1)$$

[WDT] (Appendix 2, p. 71) obtains this *empirically*: move one generator by a fixed amount, re-run the load flow, read the change in branch current. One number, one branch, one generator, one solve.

The sensitivity — what is computed here

The same quantity, signed so that positive always means *relief* of the branch it belongs to:

$$S_{e,n} = \text{PTDF}_{e,n} \cdot \text{sign}(f_e). \quad (2)$$

The sign correction is what makes a row comparable across branches: without it a positive entry means “flow increases in the branch’s reference direction”, which is relief on one branch and overload on another. With it, positive means relief everywhere, and the entries of a row may be ranked and summed.

The PTDF — what carries the network

Everything network-dependent sits in one matrix, the power transfer distribution factor:

$$\text{PTDF} = BK^{\top}L^+, \quad 980 \times 754, \quad 738,920 \text{ entries}. \quad (3)$$

PTDF $_{e,n}$ is the flow appearing on branch e when 1 MW is injected at bus n and withdrawn across the slack. It is built from topology and reactance alone — K , B and L are defined next — so it does not depend on the hour, the weather or the dispatch. Only the $\text{sign}(f_e)$ in (2) does.

Why this is available in closed form. The AC power-flow equations are nonlinear. The DC approximation linearises them under three assumptions — bus voltage magnitudes held at 1 per unit; resistance negligible beside reactance, $r \ll x$, so losses vanish; and small angle differences, so $\sin(\theta_i - \theta_j) \approx \theta_i - \theta_j$ — which makes the flow exactly linear in the angle difference:

$$f_e = \frac{\theta_{i(e)} - \theta_{j(e)}}{x_e} = b_e (\theta_i - \theta_j), \quad b_e \equiv 1/x_e. \quad (4)$$

Linearity is the whole reason a closed-form sensitivity exists. Where (1) needs one perturb-and-re-solve per generator, (3) returns every branch–node pair at once from a single inversion, exact to the model rather than estimated.

What is given up. No reactive power: real thermal limits are current limits and current tracks $|S| = \sqrt{P^2 + Q^2}$, while DC sees only P . And no voltage-stability representation at all — several real EirGrid constraint groups exist for exactly that reason and are outside anything computed here.

Every matrix that follows is from the real all-island network: 754 buses, 980 branches (755 lines and 225 transformers), 425 wind farms at 157 buses. Nothing is a reduced example.

2 Stage 1 — build K , B , φ , L

Under DC the grid is an ordinary resistive network and all three matrices are plain nodal analysis:

circuit theory	DC power flow
node voltage	bus voltage angle θ
conductance $G = 1/R$	susceptance $b = 1/x$ (this is B)
injected current source	net power injection p (MW)
nodal conductance matrix $\mathbf{G}$	$L = KBK^\top$
$\mathbf{G}V = I$ (KCL at every node)	$L\theta = p$
Ohm's law on a branch	$f = BK^\top\theta$

K , the bus–branch incidence matrix 754×980

Column e has exactly two non-zeros: $+1$ in the row of the branch's **bus0**, -1 in the row of its **bus1**.

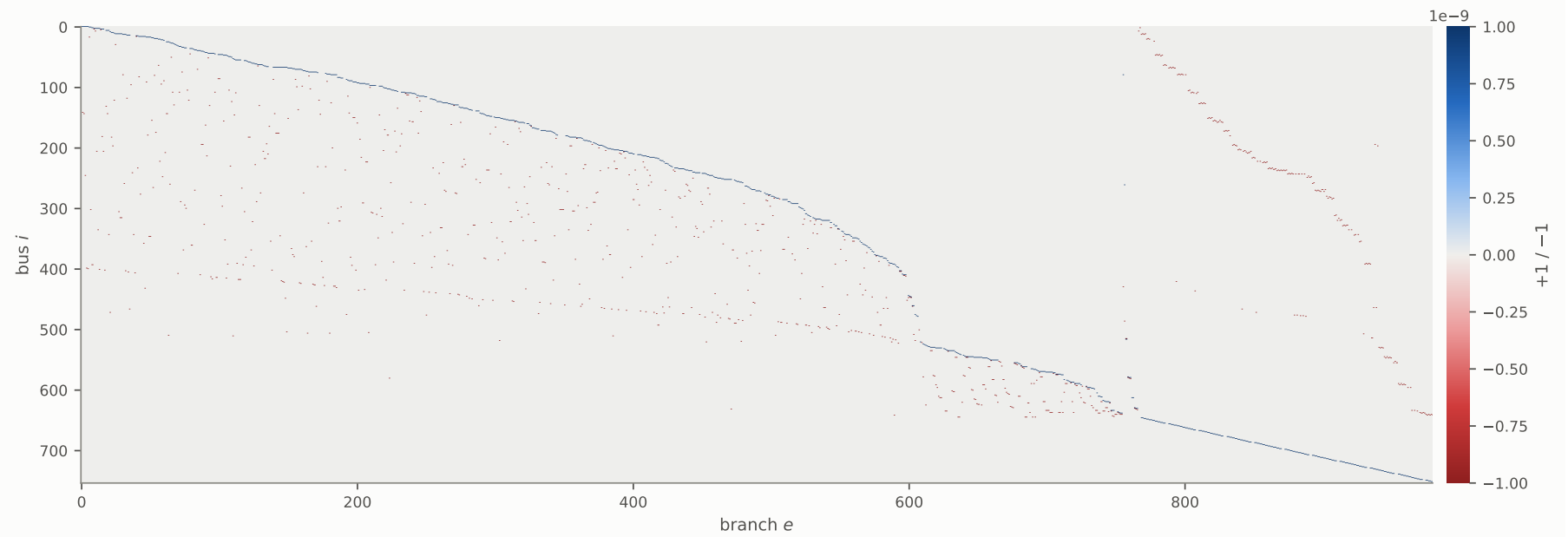
$$K_{ie} = \begin{cases} +1 & i = \text{bus0}(e) \\ -1 & i = \text{bus1}(e) \\ 0 & \text{otherwise.} \end{cases}$$

This is topology only — which buses connect to which — and carries no electrical information. It is 0.27% dense.

The orientation is arbitrary. Which end is called **bus0** was decided by row order in the data file. Harmless through Stage 3; Stage 4 exists to clean it up.

K — the full 754×980 incidence matrix

Exactly two non-zeros per column: +1 at bus0, -1 at bus1. 1,960 non-zeros, 0.27% dense.



K — real entries

10 buses against the 6 congested branches. Every column shows the same $+1/-1$ pair.

bus / branch	1122-1742-1	1122-11260-1	2781-4951-1	3082-3122-1	3581-89516-1	3691-4041-1
ARKLOW	1	1	0	0	0	0
CARRICKMINES	-1	0	0	0	0	0
GALWAY	0	0	1	0	0	0
INCHICORE	0	0	0	1	0	0
IRISHTOWN	0	0	0	-1	0	0
LETTERKENNY	0	0	0	0	1	0
LAGHTANVACK	0	0	0	0	0	1
MOY	0	0	0	0	0	-1
SALTHILL	0	0	-1	0	0	0
GLENART	0	-1	0	0	0	0

B , the susceptance matrix 980 $\times$ 980 diagonal

$$B = \text{diag}(b_1, \dots, b_{980}), \quad b_e = 1/x_e$$

in MW per radian, ranging from -21638.3 to 1754386.0 across the network. For the congested branches:

$$B_{\text{cong}} = \text{diag}(2192.0, 58823.5, 37092.0, 21285.7, 1307.2, 1109.9)$$

PyPSA stores per-unit reactance on a 1 MVA base, so $1/x_{\text{pu}}$ is already in MW/rad — verified directly: for line 1021-2121-1, $x = 7.634011 \Omega$ at 110 kV gives $x_{\text{pu}} \cdot v_{\text{nom}}^2/x = 1.000$ MVA exactly.

φ , the phase shifts 980 vector, radians

Zero for every line and every ordinary transformer. Non-zero for exactly 2 phase-shifting transformers, the larger at 17° . A shifter imposes an angle of its own, so its flow is $f_e = b_e(\theta_i - \theta_j - \varphi_e)$. It does not enter until Stage 4.

$L = KBK^T$, the **weighted Laplacian** 754×754

$$L_{ij} = \begin{cases} \sum_{e \ni i} b_e & i = j \\ -b_e & i \neq j, \text{ branch } e \text{ joins } i, j \\ 0 & \text{otherwise} \end{cases}$$

This is the DC bus admittance matrix. Applying it to the angle vector gives

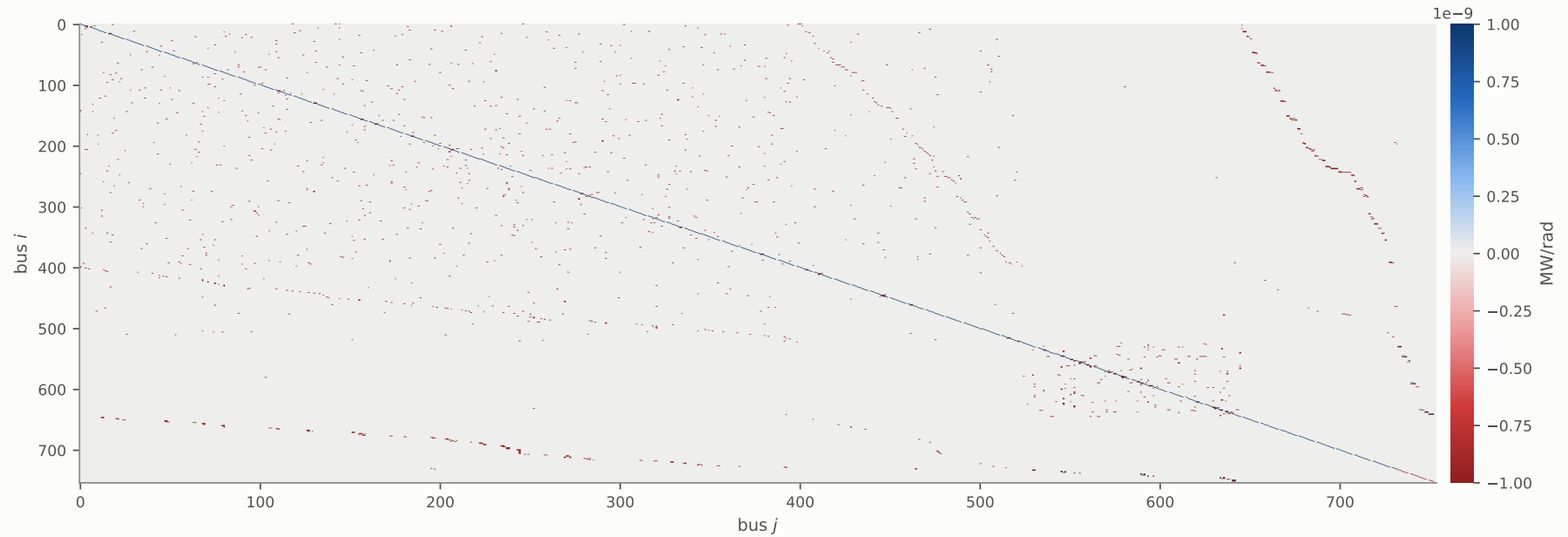
$$L\theta = p, \tag{5}$$

which is **Kirchhoff's Current Law written at every bus at once**: net power injected at a bus equals the power flowing out along its branches.

L is 0.46% dense — it inherits the grid's own sparsity, because $L_{ij} \neq 0$ only for buses that are physically connected.

$L = KBK^T$ — the full 754×754 Laplacian

2,621 non-zeros (0.46% dense). The structure is the grid's own topology.



L — real entries $(\times 10^3 \text{ MW/rad})$

	ARKLOW	CARRICKMINES	GALWAY	INCHICORE	IRISHTOWN	LETTERKENNY	LAGHTANVACK	MOY	SALTHILL	GLENART
ARKLOW	67.93	−2.19	0	0	0	0	0	0	0	−58.82
CARRICKMINES	−2.19	1093.29	0	0	−21.59	0	0	0	0	0
GALWAY	0	0	47.98	0	0	0	0	0	−37.09	0
INCHICORE	0	0	0	1035.67	−21.29	0	0	0	0	0
IRISHTOWN	0	−21.59	0	−21.29	209.55	0	0	0	0	0
LETTERKENNY	0	0	0	0	0	1019.00	0	0	0	0
LAGHTANVACK	0	0	0	0	0	0	42.78	−1.11	0	0
MOY	0	0	0	0	0	0	−1.11	2011.92	0	0
SALTHILL	0	0	−37.09	0	0	0	0	0	40.10	0
GLENART	−58.82	0	0	0	0	0	0	0	0	105.34

Diagonal entries sum over *all* branches at that bus, including ones leaving this window — so a diagonal here can exceed the off-diagonals shown. That is the one thing a window cannot show, and is why the heatmap above accompanies it.

Scale invariance

Claim. PTDF is unchanged if every susceptance is multiplied by a constant c .

Proof. Let $B \rightarrow cB$. Then $L \rightarrow cL$, and $L^+ \rightarrow L^+/c$ by the Moore–Penrose property of scalar multiples, so

$$\text{PTDF} = (cB)K^\top \left(\frac{1}{c}L^+\right) = BK^\top L^+. \quad \blacksquare$$

So the PTDF needs no MVA base. **This does not extend to Stage 4:** there $p_{\text{eff}} = p + KB\varphi$ adds a B -weighted term to a vector in MW, so once $\varphi \neq 0$, B must be in genuine MW/radian.

3 Stage 2 — the Moore–Penrose pseudoinverse

Why L cannot be inverted

Every column of K has one +1 and one −1, so

$$K^T \mathbf{1} = \mathbf{0} \implies L\mathbf{1} = KB(K^T \mathbf{1}) = \mathbf{0}. \quad (6)$$

The all-ones vector is in the nullspace: adding the same angle to every bus changes no flow. Measured, $\|L\mathbf{1}\|_\infty = 2.3e - 10$.

Here $\text{rank}(L) = 750$ of 754, so the nullity is 4 — one dimension per connected component. The extra three are the AC-isolated far terminals of the DC links (Moyle’s Scotland end, GB_EWIC, GB_GREENLINK); a DC link is a controllable injection, not a branch, so it never enters K .

What Moore–Penrose gives

L^+ is the unique matrix satisfying Penrose’s four conditions. What matters here is that it inverts L on the orthogonal complement of the nullspace and returns the **minimum-norm** solution — so $\theta = L^+p$ satisfies $\mathbf{1}^T \theta = 0$.

This is not a workaround for the singularity; it is a choice of reference, made explicit. The textbook alternative deletes a slack row and column, which writes an arbitrary choice into every number that follows.

Why it amounts to uniform distributed slack

Inject 1 MW at bus n and withdraw it evenly everywhere: $p = e_n - \frac{1}{N}\mathbf{1}$. Since $L^+\mathbf{1} = \mathbf{0}$,

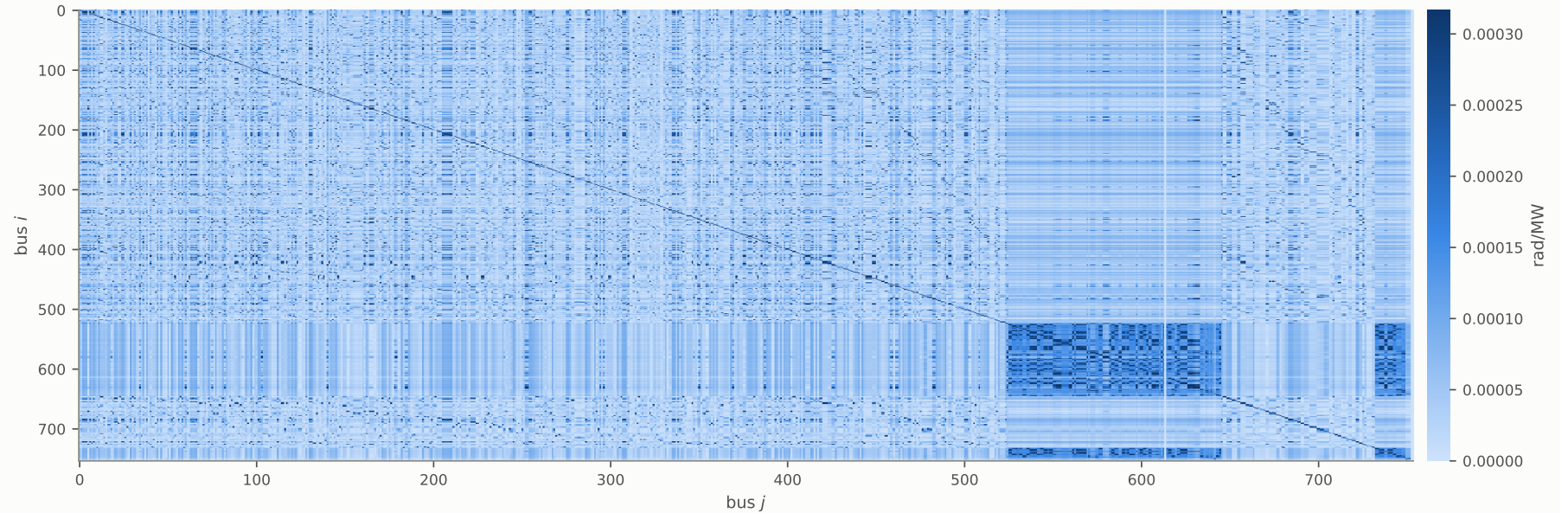
$$L^+p = L^+e_n - \frac{1}{N}L^+\mathbf{1} = L^+e_n. \quad (7)$$

So reading column n of L^+ is injecting at n against a uniform distributed slack. □

L^+ — the whole thing, and why it is expensive

L^+ — the full 754×754 pseudoinverse

99.2% dense. Inversion destroys sparsity: every bus is now coupled to every other.



L is 0.46% dense. L^+ is 99.2% dense. **Inversion destroys sparsity** — every bus becomes coupled to every other — which is why this single SVD is the only expensive step in the pipeline. It is done once and reused for every branch, every wind node and every snapshot.

L^+ — **real entries** $(\times 10^{-4} \text{ rad/MW})$

	ARKLOW	CARRICKMINES	GALWAY	INCHICORE	IRISHTOWN	LETTERKENNY	LAGHTANVACK	MOY	SALTHILL	GLENART
ARKLOW	3.386	0.896	−0.301	0.568	0.732	−0.745	−0.590	−0.594	−0.303	3.385
CARRICKMINES	0.896	1.372	−0.232	0.930	1.152	−0.585	−0.482	−0.474	−0.234	0.895
GALWAY	−0.301	−0.232	4.364	−0.220	−0.227	−0.265	1.341	0.893	4.321	−0.302
INCHICORE	0.568	0.930	−0.220	1.109	1.018	−0.552	−0.459	−0.448	−0.221	0.568
IRISHTOWN	0.732	1.152	−0.227	1.018	1.318	−0.569	−0.471	−0.462	−0.229	0.732
LETTERKENNY	−0.745	−0.585	−0.265	−0.552	−0.569	8.884	0.574	0.937	−0.267	−0.745
LAGHTANVACK	−0.590	−0.482	1.341	−0.459	−0.471	0.574	12.671	7.643	1.339	−0.591
MOY	−0.594	−0.474	0.893	−0.448	−0.462	0.937	7.643	10.029	0.891	−0.595
SALTHILL	−0.303	−0.234	4.321	−0.221	−0.229	−0.267	1.339	0.891	4.540	−0.303
GLENART	3.385	0.895	−0.302	0.568	0.732	−0.745	−0.591	−0.595	−0.303	3.555

4 Stage 3 — the PTDF

$$\boxed{\text{PTDF} = BK^{\top}L^+} \quad \underbrace{(980 \times 980)}_B \underbrace{(980 \times 754)}_{K^{\top}} \underbrace{(754 \times 754)}_{L^+} \longrightarrow 980 \times 754 \quad (8)$$

Read as two composed steps:

$$\text{PTDF} = \underbrace{BK^{\top}}_{\text{angles} \rightarrow \text{flows}} \cdot \underbrace{L^+}_{\text{injections} \rightarrow \text{angles}}$$

L^+ solves the network; $BK^{\top}$ reads flows off the angles, since $K^{\top}\theta$ is the angle difference across each branch and multiplying by $1/x$ is Ohm's law. Pre-multiplying them means the network is never solved again.

What one entry means

$$\text{PTDF}_{en} = \frac{\partial f_e}{\partial p_n} = \begin{array}{l} \text{change in flow on branch } e, \text{ in MW, per MW injected at bus } n, \text{ with the} \\ \text{balancing withdrawal spread uniformly.} \end{array}$$

Dimensionless — MW per MW. Here it spans $[-1.2093, 1.0105]$. **It is the exact analytic derivative of the model, not a finite-difference estimate;** every one of the 738,920 entries falls out of the single inversion in Stage 2.

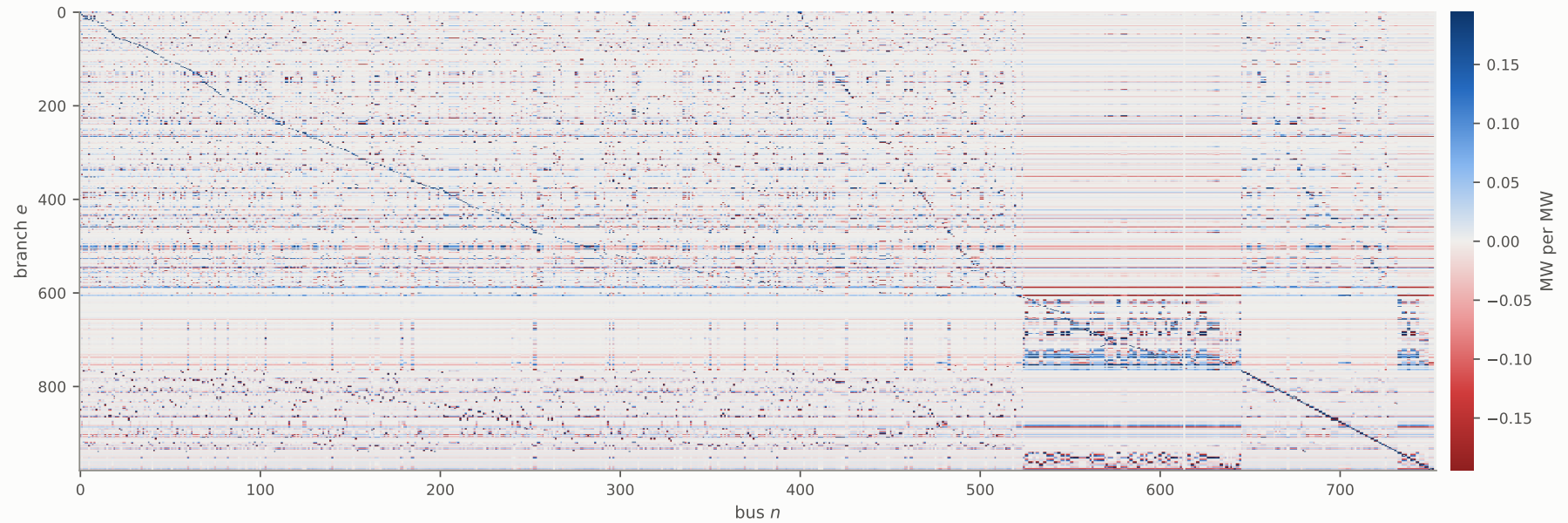
The check that costs nothing

$$\sum_n \text{PTDF}_{en} = 0 \quad \text{for every } e, \quad (9)$$

because injecting 1 MW at every bus simultaneously moves no power. Measured: $3.6e - 10$.

PTDF = BK^TL^+ — all $980 \times 754 = 738,920$ entries

Every branch against every bus. No numeric form of this is readable, so it is shown whole as an image.



PTDF — real entries

The 6 congested branches against ten buses. A star marks a bus with wind on it; the rest are load, conventional generation, or bare junctions.

branch / bus	ARDNACRUSHA*	BALRUNTAGH	ARIGNA*	AGHADA	ARKLOW*	AGHADA	ATHEA*	AGHADA_B	ATHLONE*	AGH_LAT14
1122-1742-1	0.0187	-0.0406	-0.0295	0.0707	0.4516	0.0706	0.0004	0.0706	-0.0109	0.0706
1122-11260-1	0.0027	0.0027	0.0027	0.0027	0.0027	0.0027	0.0027	0.0027	0.0027	0.0027
2781-4951-1	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
3082-3122-1	-0.0128	0.0494	0.0335	-0.0519	-0.3496	-0.0518	-0.0050	-0.0518	0.0186	-0.0518
3581-89516-1	-0.0024	0.0146	0.0919	-0.0054	-0.0097	-0.0055	-0.0056	-0.0055	0.0135	-0.0055
3691-4041-1	0.0114	-0.0231	-0.0694	0.0061	0.0005	0.0060	0.0064	0.0060	0.0035	0.0060

Both signs occur and both are meaningful. A negative entry means injection at n pushes flow in the `bus1`→`bus0` direction — it does *not* yet mean “makes congestion worse”. That reading needs Stage 4.

5 Selecting the wind nodes

PTDF has a column for every one of the 754 buses. Only 157 of them carry wind, and only wind can be curtailed — so the 597 remaining columns are not part of any curtailment decision.

This is a column selection and nothing more. Write S for the 754×157 matrix with a single 1 in each column, at the row of that wind bus:

$$\text{PTDF}_{\text{wind}} = \text{PTDF} \cdot S, \quad S \in \{0, 1\}^{754 \times 157}, \quad 980 \times 754 \longrightarrow 980 \times 157. \quad (10)$$

No arithmetic happens. Nothing is recomputed. In code it is `ptdf[:, wind_buses]`.

Why the other 597 buses cannot be dropped earlier

They are not padding — they are the network the power flows *through*. Column n of PTDF is $BK^T(L^+e_n)$, and L^+e_n is the whole network's response to an injection at n . That response depends on every bus and every branch. **There is no 157×157 Laplacian to build:** delete the Dublin load buses and the Donegal wind columns change.

So the selection has to come *after* Stages 1–3, never before. One could compute only the 157 needed columns of L^+ — that is 157 least-squares solves rather than a full pseudoinverse — but on a 754×754 matrix the full `pinv` takes under a second, and computing all of it buys the row-sum check across every bus for free.

Kron reduction onto the wind buses would also be wrong here. It preserves behaviour at the retained buses but discards flows on the eliminated branches — and every branch is monitored, including the 0 congested transformers.

Selection and the sign correction commute

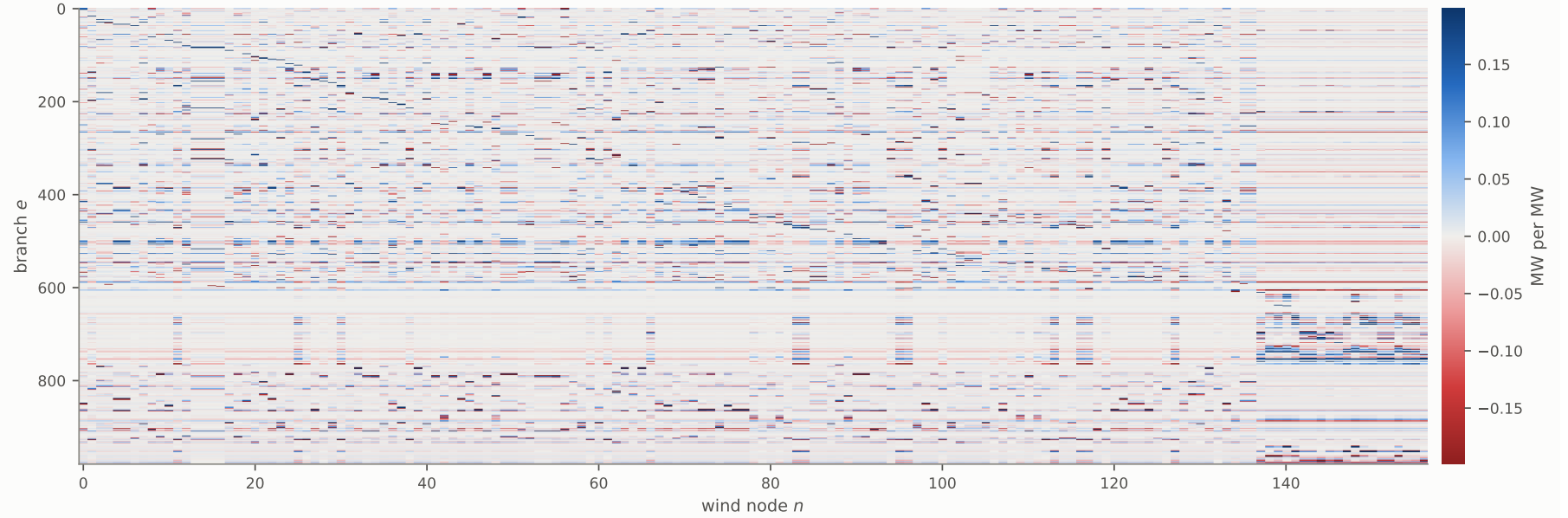
The sign correction of Stage 4 scales each *row*; the selection keeps certain *columns*. Row scaling and column selection act on different indices, so

$$(\text{diag}(\sigma) \text{PTDF})S = \text{diag}(\sigma) (\text{PTDF} S), \quad \sigma_e = \text{sign}(f_e). \quad (11)$$

Verified numerically: the two orderings differ by $0.0e + 00$. The code slices first and signs second; this document signs first and slices second. Same matrix either way.

PTDF · S — the same matrix, 157 wind columns kept

A column selection, nothing more: 980×157 . The other 597 buses were needed to compute these values and are not curtailable.



PTDF · S — real entries

The same 6 congested branches, now against wind nodes only:

branch / wind bus	ARDNACRUSHA	ARIGNA	ARKLOW	W_ARKLOW_OFF	ATHEA	TOBERTOREEN	ATHLONE	BALLYWATER	BOOLTIAGH	BALLYLICKEY
1122-1742-1	0.0187	-0.0295	0.4516	0.5458	0.0004	0.0004	-0.0109	0.3563	0.0026	0.0559
1122-11260-1	0.0027	0.0027	0.0027	-0.9973	0.0027	0.0027	0.0027	0.0027	0.0027	0.0027
2781-4951-1	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
3082-3122-1	-0.0128	0.0335	-0.3496	-0.3495	-0.0050	-0.0050	0.0186	-0.2528	-0.0027	-0.0419
3581-89516-1	-0.0024	0.0919	-0.0097	-0.0097	-0.0056	-0.0056	0.0135	-0.0088	-0.0015	-0.0054
3691-4041-1	0.0114	-0.0694	0.0005	0.0005	0.0064	0.0064	0.0035	0.0015	0.0137	0.0062

6 Stage 4 — correct the sign direction

The problem

Two quantities sit on the *same arbitrary axis* set by the `bus0/ bus1` labelling: the base flow f_e , and the row $\text{PTDF}_{e,:}$. That axis came from data-file row order.

Congestion, however, is about *magnitude* — $|f|$ against the rating:

base flow on e	overload means	relief requires
positive (bus0→bus1)	f too large positive	$\text{PTDF}_{en} > 0$
negative (bus1→bus0)	f too large negative	$\text{PTDF}_{en} < 0$

So the sign meaning “relieves congestion” flips with an arbitrary labelling choice. Roughly half of all branches are affected, and a sign-flipped ranking looks entirely plausible on inspection.

Getting f right: the phase-shift term

A phase shifter imposes its own angle, so $f = B(K^\top\theta - \varphi)$. KCL still gives $Kf = p$, so

$$KB(K^\top\theta - \varphi) = p \tag{12}$$

$$L\theta = p + KB\varphi. \tag{13}$$

The shifter appears exactly where an injection would — **it behaves as a pair of equal and opposite injections at its two ends.** Define $p_{\text{eff}} = p + KB\varphi$; then

$$\boxed{\theta = L^+ p_{\text{eff}}, \quad f = BK^\top\theta - B\varphi = \text{PTDF} \cdot p_{\text{eff}} - B\varphi} \tag{14}$$

Same L^+ , same PTDF, one modified input vector and one constant subtraction.

Three properties. (i) $\mathbf{1}^\top p_{\text{eff}} = \mathbf{1}^\top p + (K^\top \mathbf{1})^\top B\varphi = 0$, so balance survives — measured $1.3e - 12$. (ii) Expanding, $f = \text{PTDF} \cdot p + c$ with $c = (BK^\top L^+ K - I)B\varphi$ independent of p : a constant offset, not a different model. (iii) At $\varphi = 0$ it reduces term by term to $f = \text{PTDF} \cdot p$.

PTDF is untouched by any of this: $\partial f / \partial p = \text{PTDF} \cdot \partial p_{\text{eff}} / \partial p = \text{PTDF}$, because φ is a fixed device setting, not a function of injection.

Why the term is not optional

This network has 2 phase shifters, one at 17°. Dropping $B\varphi$ leaves the most heavily loaded circuit in the network, 3581-89516-1, reading 44.2 MW against a 123 MVA rating — 36% loaded — when it is in fact at 123.0 MW, **exactly at its limit**. It stops looking congested at all. Across the network, 272 of 980 branches come out wrong by more than 1 MW, worst case 78.8 MW.

The fix

$$\boxed{\text{sensitivity}_{e,:} = \text{PTDF}_{e,:} \cdot \text{sign}(f_e)} \quad (15)$$

$\text{sign}(f_e) = +1$	axis already runs along the real flow; row unchanged
$\text{sign}(f_e) = -1$	axis runs backwards; row flipped

Verification against EirGrid’s convention. Let $S = \text{PTDF}_{en} \text{sign}(f_e)$. Reducing generation by $r > 0$ at n is an injection change of $-r$, so the flow-aligned quantity changes by $-Sr$:

- $S > 0$: change is negative, flow shrinks along its own direction, magnitude falls — **congestion relieved**. ✓
- $S < 0$: flow grows along its own direction, magnitude rises — **congestion worsened**. ✓

Flows and signs — real values at 2030-01-16 16:00:00

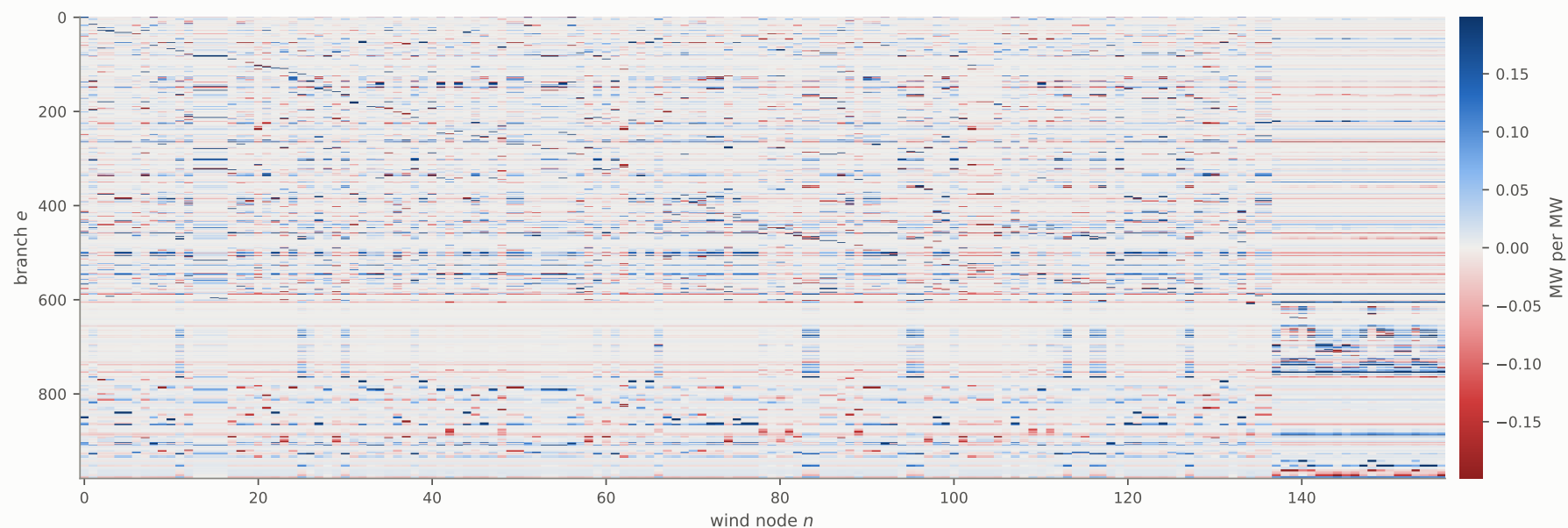
branch	f (MW)	sign	peak load %
1122-1742-1	281.79	1.00	100.00
1122-11260-1	-228.78	-1.00	100.00
2781-4951-1	-64.15	-1.00	100.00
3082-3122-1	-378.14	-1.00	100.00
3581-89516-1	123.00	1.00	100.00
3691-4041-1	81.00	1.00	100.00

This is the only snapshot-dependent step. Stages 1–3 depend on topology and reactance alone and hold for every hour of the year. Measured across all 168 snapshots, a branch’s sign agrees with its own peak-hour sign 98.6% of the time when it is above 50% loaded (87.4% at worst).

7 The result

Sensitivity — 980×157 , signed along real flow

PTDF re-pointed by $\text{sign}(f_e)$. Positive means curtailing there unloads the branch.



Sensitivity — real entries

branch / wind bus	ARDNACRUSHA	ARIGNA	ARKLOW	W_ARKLOW_OFF	ATHEA	TOBERTOREEN	ATHLONE	BALLYWATER	BOOLTIAGH	BALLYLICKEY
1122-1742-1	0.0187	-0.0295	0.4516	0.5458	0.0004	0.0004	-0.0109	0.3563	0.0026	0.0559
1122-11260-1	-0.0027	-0.0027	-0.0027	0.9973	-0.0027	-0.0027	-0.0027	-0.0027	-0.0027	-0.0027
2781-4951-1	-0.0059	-0.0059	-0.0059	-0.0059	-0.0059	-0.0059	-0.0059	-0.0059	-0.0059	-0.0059
3082-3122-1	0.0128	-0.0335	0.3496	0.3495	0.0050	0.0050	-0.0186	0.2528	0.0027	0.0419
3581-89516-1	-0.0024	0.0919	-0.0097	-0.0097	-0.0056	-0.0056	0.0135	-0.0088	-0.0015	-0.0054
3691-4041-1	0.0114	-0.0694	0.0005	0.0005	0.0064	0.0064	0.0035	0.0015	0.0137	0.0062

positive	curtailing at that wind node unloads the branch
negative	curtailing at that wind node loads it further

Of 980 branches, 6 reach their rating during the week — 0 of them transformers, which a lines-only model would miss entirely.

Summary

1. $K = \text{incidence}$, $B = \text{diag}(1/x)$, $\varphi = \text{phase shifts}$, $L = KBK^\top$
2. $L^+ = \text{pinv}(L)$
3. $\text{PTDF} = BK^\top L^+$ 980×754 , snapshot-independent
4. $p_{\text{eff}} = p + KB\varphi$, $f = \text{PTDF} \cdot p_{\text{eff}} - B\varphi$, $\text{sensitivity}_{e,:} = \text{PTDF}_{e,:} \text{sign}(f_e)$

Stages 1–3 are pure network. Stage 4 is the only part that moves hour to hour.

check	value	must be	meaning
flows vs. PyPSA <code>n.lpf()</code>	$2.1e - 09$ MW	≈ 0	an independent DC solver agrees
PTDF row sums	$3.6e - 10$	0	1 MW at every bus at once moves nothing
$\ L\mathbf{1}\ _\infty$	$2.3e - 10$	0	the constant vector is in the nullspace
$\mathbf{1}^\top p$	$1.6e - 12$	0	injections balance
$\mathbf{1}^\top p_{\text{eff}}$	$1.3e - 12$	0	the phase-shift term preserves that

Method and derivations: `shift-factor-ptdf-report.md`. Data: `results/`. Every number above is emitted by `build.py` from the same code that produced those CSVs — none is typed by hand.

Limits. DC approximation: no reactive power, no losses, no voltage, so voltage-stability constraints cannot be represented. Intact base case only — no contingencies, though real limits usually bind post-fault. Ratings are TYTFS `RATE1` planning values. All time series in the source network are synthetic, so the congested set is one draw from one modelled week; the PTDF itself does not depend on it.

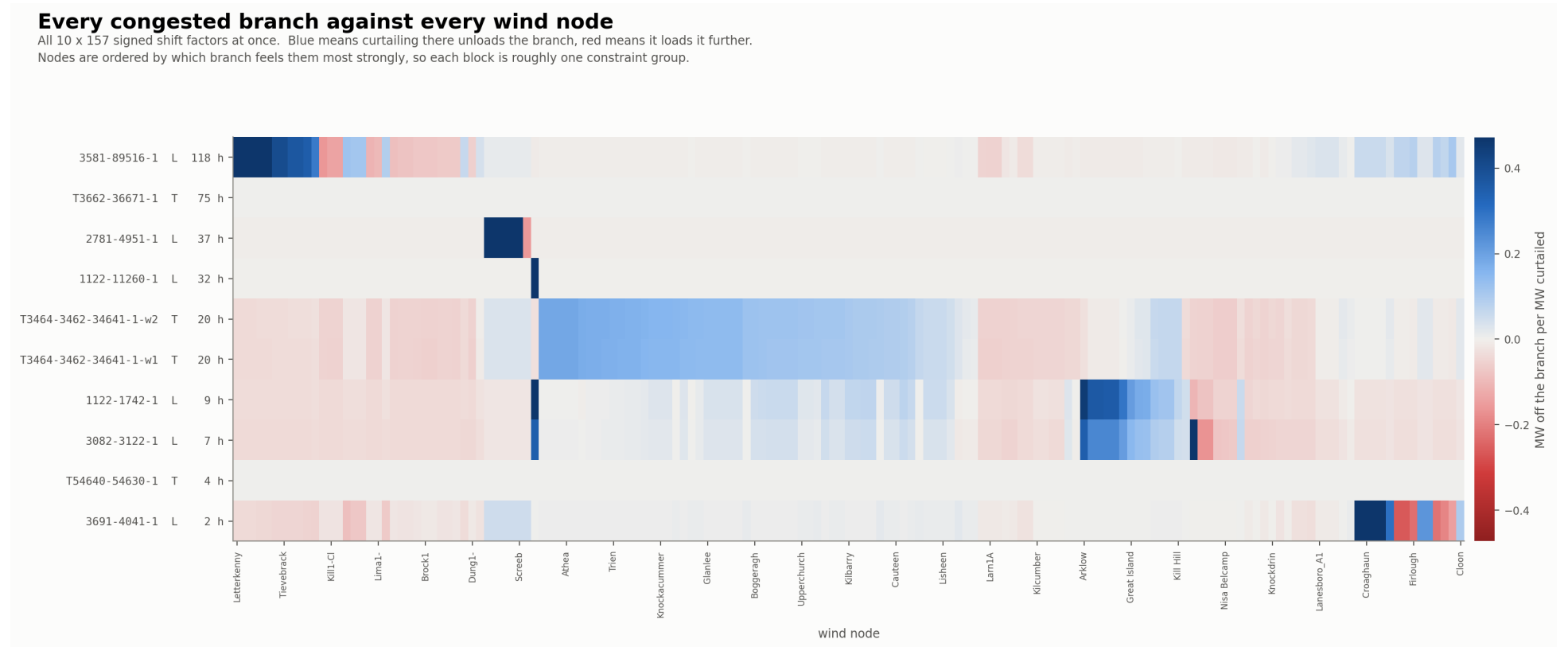
8 Net sensitivity: a constraint group that cannot break what it did not look at

Shift factor, revisited

Every quantity a constraint group is built from is already in hand: row e of the sensitivity matrix (§A, Stage 4) gives, for branch e , the change in $|f_e|$ per MW curtailed at every wind node — signed so that positive means *relief*. [WDT]’s method (Appendix 2, p.72) reads a *single* row: pick the one congested circuit, rank nodes down that row, keep whatever clears a threshold τ .

$$\text{group}(e) = \{n : \text{sensitivity}_{e,n} \geq \tau\}, \quad \tau = 0.05. \quad (16)$$

That is one row of a 980×157 matrix, chosen and read in isolation. Nothing in it depends on any other row — so nothing in it *knows* about any other row.



Every congested branch against every wind node. Each row is a constraint group as [WDT] would draw it — read left to right, in isolation. The blindness is visible: a node blue on one row is often red on another, and the method that picked it never looked at that second row.

Sensitivities add. Net sensitivity is nothing more than that.

If several branches are over their rating at once, a node's total usefulness is just its sensitivities to each of them, summed:

$$S_{\text{net}}(n) = \sum_{e \in \text{Over}} \text{sensitivity}_{e,n}, \quad \text{Over} = \{ e : |f_e| > s_e \}. \quad (17)$$

No new computation — the matrix already holds every term on the right. A node relieving three circuits a little now outranks one relieving only the worst circuit a lot; a one-line group is unaffected, since (17) collapses to the single term τ was already thresholding.

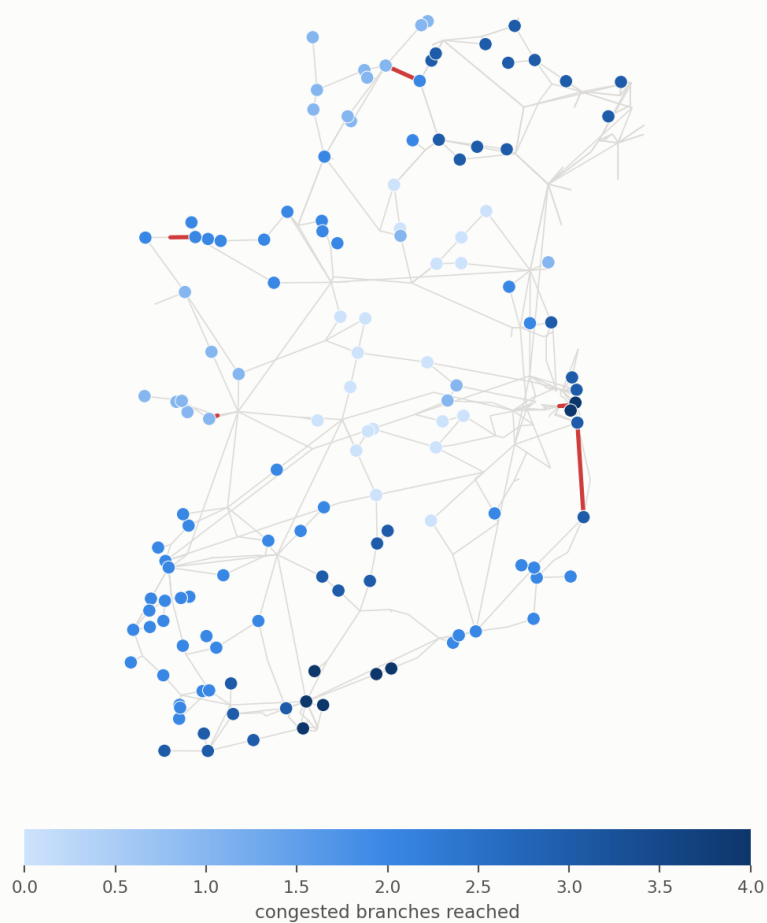
Reach and net relief, on the ground

Left: how many of the ten congested branches each wind node reaches. Right: its net weighted relief, which is what breadth is worth once the signs are counted.

A node can be dark on the left and pale on the right — broad, but pulling in both directions at once.

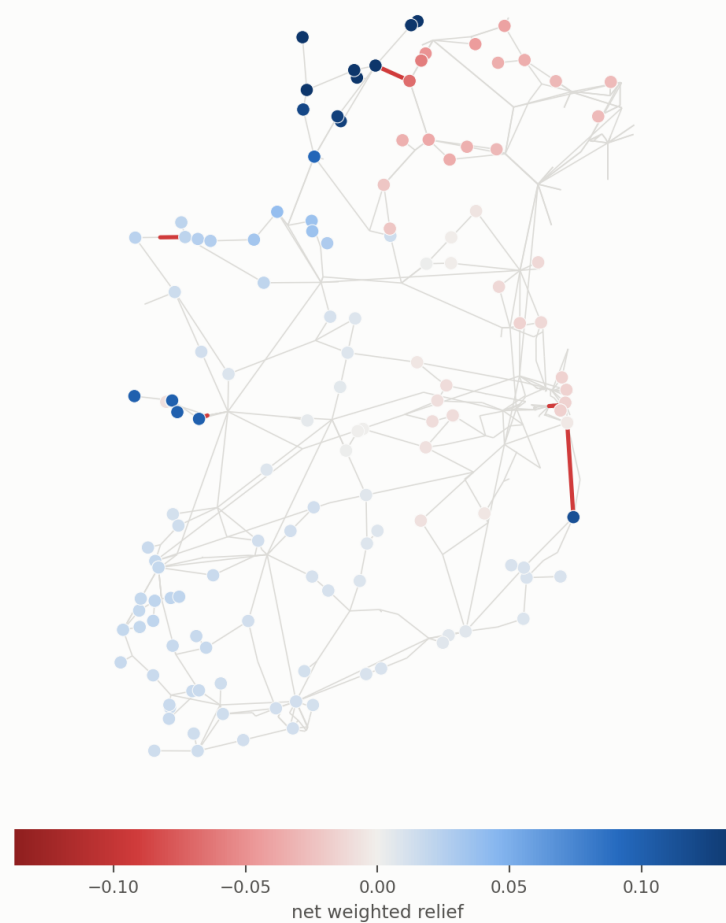
Reach

Red edges are the ten congested branches.



Net relief

Red edges are the ten congested branches.



Left: how many congested branches each wind node reaches. Right: its net weighted relief, (17), on the same map. A node dark on the left and pale on the right is broad

but pulling both ways at once — which is exactly the case counting rows cannot distinguish and summing them can.

Why net sensitivity alone is not enough

Ranking by S_{net} still asks only *”does this node help the branches I am watching.”* It says nothing about the 980 – |Over| branches it is not watching. Tested directly: ranking by S_{net} with no further check pushed a branch that was inside its rating over it in 1,085 of 20,203 events — *worse* than the threshold method it replaces. Concentrating a whole instruction on the single most useful node moves the rest of the network harder than the same megawatts spread out. **The safety was never going to come from the ranking.** It has to come from a separate rule.

The veto

A node’s cut is capped, not by τ , but by the largest amount that leaves *every live branch* k — not just the ones in Over — inside its rating, and leaves nothing already over made worse:

$$x_n^{\max} = \min_{k : \text{sensitivity}_{k,n} < 0} \frac{\text{rating}_k - |f_k|}{|\text{sensitivity}_{k,n}|}, \quad \text{lim}_k = \max(|f_k|, \text{rating}_k) \text{ for } k \in \text{Over}. \quad (18)$$

Every branch in the network enters this minimum, whether it is congested or not — because the branch a bad instruction breaks is, by definition, one that was not being watched. A node is included in the group only up to $x_n^{\max}$; ordering by S_{net} decides *who goes first*, (18) decides *how far they may go*. The two are independent checks and the results below keep them separate.

	cleared	events broke a line	lines broken	median group
Shift-factor threshold (today)	92.0%	642	873	18 nodes
Net sensitivity + veto	89.9%	0	0	1 node

20,203 (outage, hour) events, every single-branch outage at every hour of the week. On the events both methods clear, the veto method cuts 196,045 MW less wind — 23.8% — because it targets the effective node instead of splitting pro-rata across 18. The proven optimum (the same rule in (18), solved exactly rather than applied greedily) clears 99.8%; a safe remedy provably exists in 99.8% of events.

Fixed, and precomputable, per outage

Neither (17) nor (18) depends on the hour or the weather — both are read off the sensitivity matrix, which is topology and reactance only. So the group for *every* single-branch outage can be built once, offline, and looked up: 432 outages produce 58 distinct circuits ever at risk, 1,308 (outage, circuit) lookup rows in total. 5,100 of the 58-circuit threshold method’s own 18-node-median memberships (10.0% of every member ever proposed) fail (18) and are excluded before dispatch. Only the live flow and the final MW instruction are computed online.

Finding 1 — curtailment is concentrated on a handful of nodes

Running (18) across every one of the 20,203 events and tracking which nodes the safe remedy actually dispatches, not merely which ones qualify as candidates: no node is cut in *zero* events — the least-used one is still asked 6 times out of 20,203— so there is no node that can be declared exempt outright. What the same sweep shows instead is concentration. The **5** busiest nodes account for 46% of every megawatt curtailed anywhere in the study; the top 10 for 59%; the top 20 for 72%. 119 of 157 nodes — 75.8% of the fleet, 8,367 MW of nameplate, 65.9% of the 12,701 MW installed — are cut in under 1% of the 20,203 events, and the median node across the whole fleet is cut in only 0.52% of them, keeping 99.61% of whatever it was offering when it was. Overall the safe remedy cuts 0.85% of all wind offered across the entire N–1 study: 921,085 MW of 108,051,058 MW. **For the great majority of the fleet, a constraint group is something that applies to them on a handful of days a year, not a standing limit on their output.**

Finding 2 — stopping the cascade

Follow the chain past the first trip: whatever is left over its rating after the remedy trips too, and the flows are recomputed with it also out of service. 250 starting outages, run to exhaustion under four remedies:

	lines lost, total	outages that spread	islanded
No remedy	814	250 / 250	169
Shift-factor threshold	309	137 / 250	72
Net sensitivity + veto	261	119 / 250	55
Safe optimum (LP)	38	25 / 250	7

This sample is deliberately enriched with the events where a remedy was already seen to break something, so the spread rate above is not the rate for a uniformly random N–1 — it is a stress test, by design.

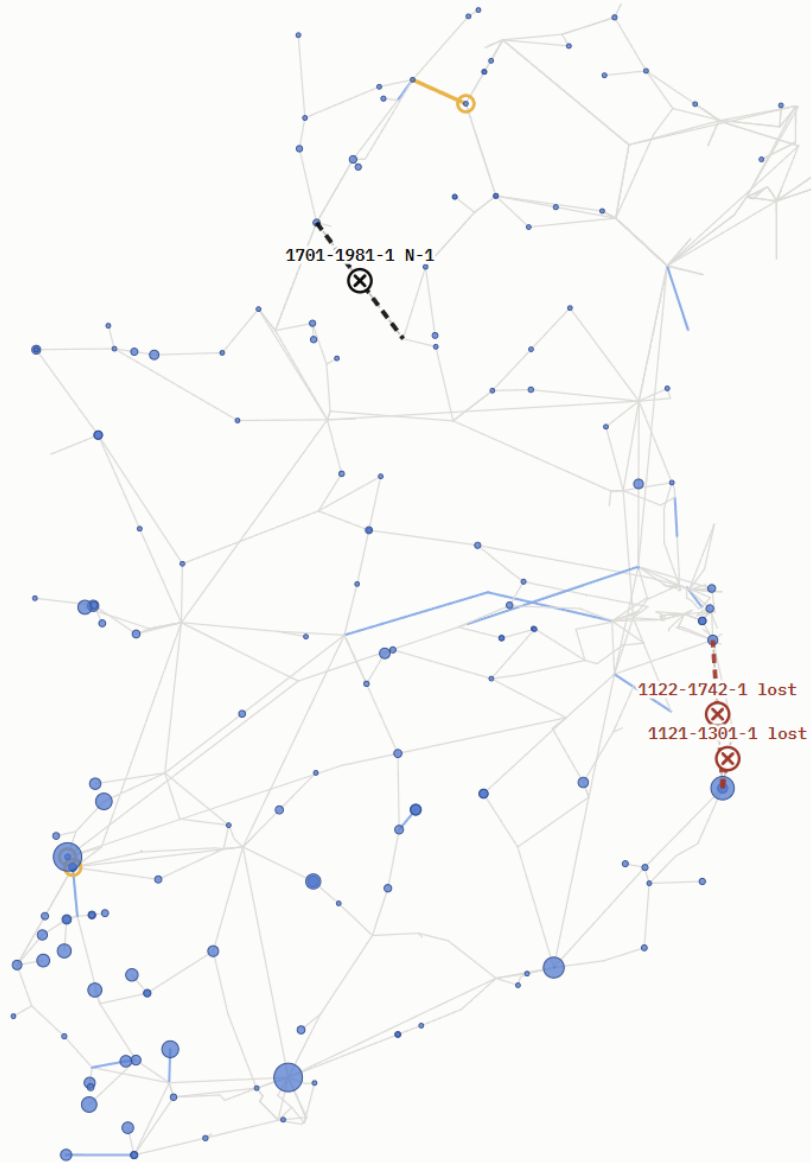
Shift factor constraint groups

LINE OUTAGES

2

WIND CUT

744 MW · 14%



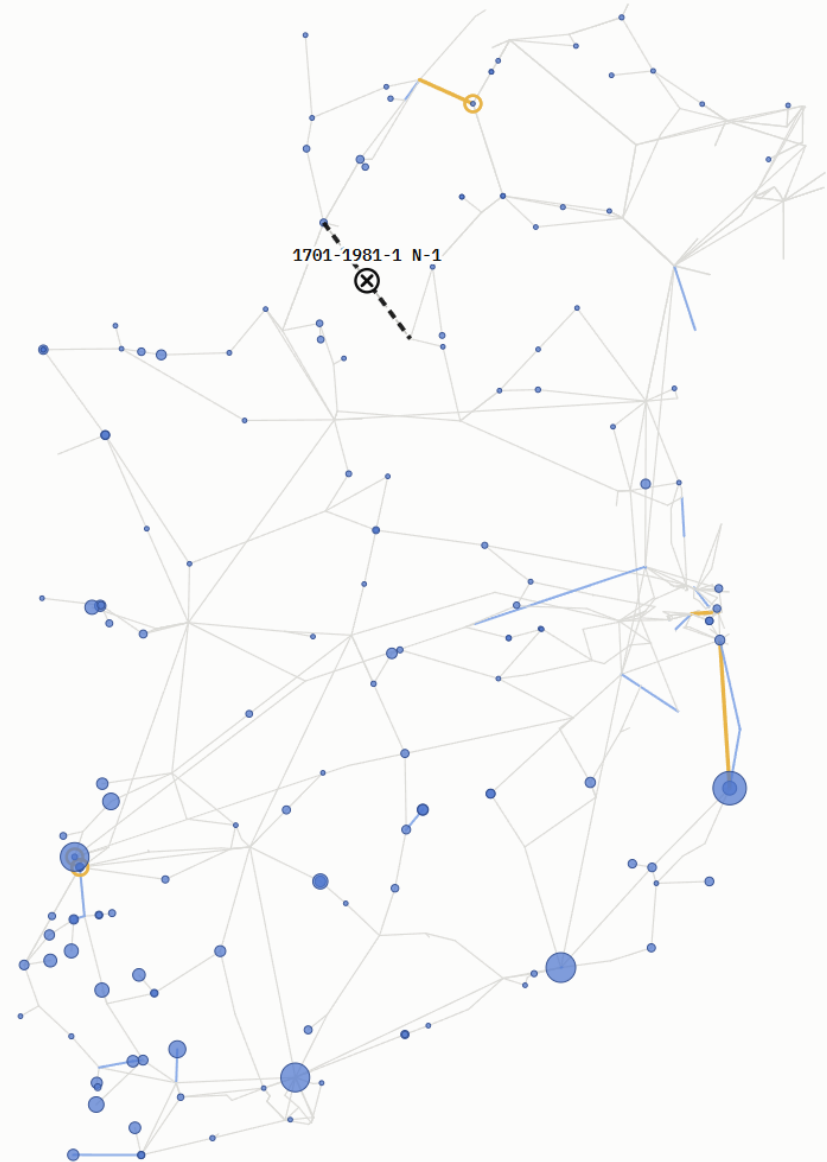
Net sensitivity constraint groups

LINE OUTAGES

0

WIND CUT

16 MW · 0%



One outage, side by side: line 1701-1981-1 trips, and 3581-89516-1 goes to 107.6% of its rating. **Left**, the shift-factor group: it curtails 744 MW — 14% of the fleet — over three rounds and still loses two lines, 1122-1742-1 then 1121-1301-1, the first of them broken by the remedy itself. **Right**, net sensitivity with the veto: 16 MW at a handful of nodes, one round, nothing lost. A $46\times$ difference in curtailment on the same fault. Row 1701-1981-1, hour 50 of `cascade_ranked.csv`; unarrested, the same fault takes 17 lines and islands the network. **Limits**. No thermal clock: a branch over its rating at the end of a round is taken to trip, the pessimistic reading, since real protection is time-overcurrent. The veto is exact only inside the DC linearisation it shares with the rest of this document. Weather is synthetic, so the event count and the MW figures move with real data; *which* circuits are ever at risk does not, since it follows from K , B and the ratings alone.

A Complete matrices, entry by entry

Not generated in this build. Run

```
python build.py --full-dump && latexmk -pdf pipeline.tex
```

to include every entry of K , L , L^+ , PTDF and the sensitivity matrix, split into column blocks narrow enough for $\text{\TeX}$ to typeset. It is off by default because it runs to several hundred pages: L^+ alone is 568,516 entries and PTDF is 738,920.

The same numbers are available in a form better suited to reading: `results/ptdf/ptdf_wind.csv`, `results/sensitivity/sensitivity_wind.csv`, and the interactive `results/sensitivity/viewers/ptdf_viewer_withOverlaps.html`.